

ORF397 Optimization

2025 Fall ORF307 Final Exam

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Instructions

- Exam files: the exam problems (as a PDF) are available to download at Gradescope. Starter Python code and Notebook are available on Canvas → Assignments → Final Exam.
- Date and time: From December 13, 7:00 AM to December 14, 7:00 PM.
- Write your full name and honor code at the top of your solutions.
- Write your answers clearly and legibly.

Exam Rules

- You are allowed to use all course materials on the final exam (lecture notes, books, precept materials, code, and homeworks). You cannot use internet to search for answers directly related to the problem in the exam.
- Strictly Prohibited: The use of any AI-assisted tool (e.g., ChatGPT, Gemini, Claude, etc.) is strictly prohibited.
- You have to justify all your answers to get full credit. If you use code from the course materials, you have to explain what each step means.
- The output and results you present must be verifiable from your included code.
- You cannot communicate with anyone during the exam.
- No late submissions allowed. Make sure your submission goes through on time. You can resubmit as many times as you like until your time expires.
- The exam must be submitted electronically as a single PDF on Gradescope before the deadline December 14, 7:00 PM.

Note: Problem 1 and Problem 2 are related and must be solved in sequence. Problem 3 is independent of the other.

1. *Problem 1 (30 points). Rural Lifeline Logistics - Optimal Redistribution*

Rural Lifeline Logistics operates six (6) regional centers and needs to efficiently **redistribute** a life-saving medicine among all centers. For each center, the company knows the units that are available (b_i) and the required demand (d_i). The corresponding data is shown in Table 1.

Center i	Units Available b_i	Units Required d_i
1	10	15
2	20	10
3	5	10
4	15	20
5	10	5
6	15	15

Table 1

The transportation network and corresponding unit costs are summarized in Table 2. For example, transporting one unit of medicine from center 1 to center 2 costs \$2.5. The main goal of the company is to determine the optimal number of units to transport between centers to satisfy all required demand while minimizing the total transportation cost.

Arc (i, j)	Cost (\$) c_{ij}	Arc (i, j)	Cost (\$) c_{ij}
(1, 2)	2.50	(3, 5)	2.00
(2, 4)	1.80	(4, 5)	0.90
(2, 6)	3.10	(5, 1)	4.00
(3, 4)	1.20	(6, 3)	1.50

Table 2

- (5 points) For each center, compute the net flow requirement $R_i = b_i - d_i$. Based on the values you find clearly categorize each center as a supply node, a demand node, or transshipment node.
- (5 points) Formulate the company's problem as a Minimum Cost Flow LP problem. Clearly state and explain the decision variables, objective, and flow conservation constraints.
- (5 points) Use **CVXPY** to solve the LP problem. Report the optimal total cost (Z^*) and list the optimal flows (x_{ij}) for all arcs.
- (5 points) Explain why the optimal solution consists of integer values (no need to prove it).
- (5 points) Suppose that due to a car accident, we can only transport at most 4 units of medicine along arc (2, 4) (perhaps due to closure of some lanes). Modify your original **CVXPY** code from part (c) to include this new constraint.
 - Report the new optimal total cost Z^{**} .
 - List the new set of optimal flows.
 - Explain the main reason the total cost changed.
- (5 points) Define the Dual of the Minimum Cost Flow LP problem in part (b), and solve it using **CVXPY**.
 - Report the dual optimal value Ω^*
 - List the values of the optimal dual variables.
 - Briefly discuss their economic interpretation.

Solution:

We first note that the transportation network that connects the centers can be defined by the graph $G = (V, E)$ where

$$V = \{1, 2, 3, 4, 5, 6\}$$

is the set of nodes and

$$E = \{(1, 2), (2, 4), (2, 6), (3, 4), (3, 5), (4, 5), (5, 1), (6, 3)\}$$

is the set of arcs (edges).

The network is shown in Figure 1.

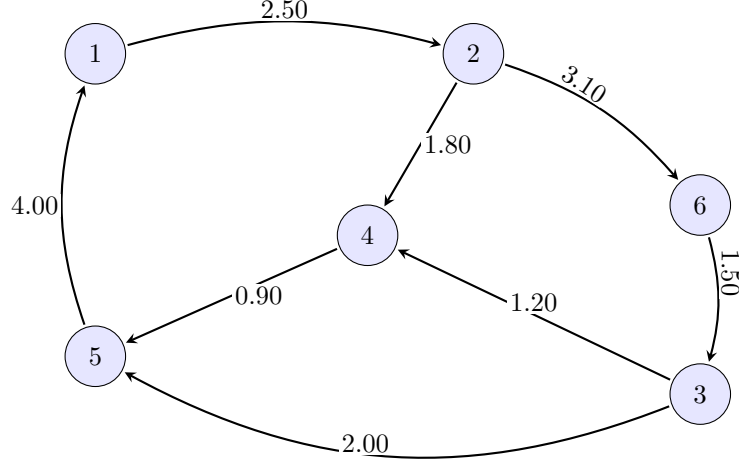


Figure 1: The Minimum Cost Flow network for the logistics problem. Arc weights represent the unit transportation cost c_{ij} .

- (a) The difference between the available units, b_i , and the required demand, d_i , will give us the net-flow for each center i , that is, $R_i = b_i - d_i, i = 1, 2, 3, 4, 5, 6$. In particular, we have
- Center 1: $R_1 = 10 - 15 = -5$
 - Center 2: $R_2 = 20 - 10 = +10$
 - Center 3: $R_3 = 5 - 10 = -5$
 - Center 4: $R_4 = 15 - 20 = -5$
 - Center 5: $R_5 = 10 - 5 = +5$
 - Center 6: $R_6 = 15 - 15 = 0$.

The vector that contains the net flows for all centers is

$$R = b - d = (-5, 10 - 5, -5, 5, 0)^T.$$

Based on the definition of the net flow vector R we have the following types of nodes

- Supply nodes ($R_i > 0$): center 2, center 5
- Demand nodes ($R_i < 0$): center 1, center 3, center 4
- Transient nodes ($R_i = 0$): center 6.

We note that the total availability (75) is equal to the total demand(75) and hence the problem is feasible.

- (b) We first define the decision variables, $x_{ij}, \forall (i, j) \in E$, which represent the number of units moved from center i to center j (i.e., on the arc (i, j)). Note that we only define decision variables for those $(i, j) \in E$.

Next we define the objective function, which is the total transportation cost

$$Z = \min \quad 2.5x_{12} + 1.8x_{24} + 3.1x_{26} + 1.2x_{34} + 2.0x_{35} + 0.9x_{45} + 4.0x_{51} + 1.5x_{63}$$

For the constraints we have the non-negativity of all decision variables, i.e., $x_{ij} \geq 0, \forall (i, j) \in E$,

and the balance (or flow conservation) constraints

$$\begin{aligned}
\text{Demand node 1} &\rightarrow x_{12} - x_{51} = -5 \\
\text{Supply node 2} &\rightarrow x_{24} + x_{26} - x_{12} = +10 \\
\text{Demand node 3} &\rightarrow x_{34} + x_{35} - x_{63} = -5 \\
\text{Demand node 4} &\rightarrow x_{45} - x_{24} - x_{34} = -5 \\
\text{Supply node 5} &\rightarrow x_{51} - x_{35} - x_{45} = +5 \\
\text{Transshipment node 6} &\rightarrow x_{63} - x_{26} = 0 \\
\text{Non-negative flows} &\rightarrow x_{ij} \geq 0, \forall (i, j) \in E
\end{aligned}$$

(c) Your Python code should produce the following results.

The minimum total transportation cost is $Z^* = 52$ and the optimal flows $x_{ij}, \forall (i, j) \in E$ are given by

- $x_{12} = 0.0$
- $x_{24} = 5.0$
- $x_{26} = 5.0$
- $x_{34} = 0.0$
- $x_{35} = 0.0$
- $x_{45} = 0.0$
- $x_{51} = 5.0$
- $x_{63} = 5.0$

(d) Total Unimodularity is used to deduce that the solution of the Minimum Cost Network Flow (MCNF) problem has an integer solution. Indeed, our original problem is, strictly speaking, an Integer Programming problem (we can only send whole packets of medicine across the arcs of the transportation network). But since the constraint matrix A contains only one $-1, +1, 0$ in each of its columns, and the right-hand side of the constraints contains an integer vector, the solution of the MCNF problem, considered as an LP, is always integer.

(e) The new constraint is described by the inequality: $x_{24} \leq 4$, and the new MCNF problem is defined as follows

$$\begin{aligned}
\min \quad & 2.5x_{12} + 1.8x_{24} + 3.1x_{26} + 1.2x_{34} + 2.0x_{35} + 0.9x_{45} + 4.0x_{51} + 1.5x_{63} \\
s.t. \quad & x_{12} - x_{51} = -5 \\
& x_{24} + x_{26} - x_{12} = +10 \\
& x_{34} + x_{35} - x_{63} = -5 \\
& x_{45} - x_{24} - x_{34} = -5 \\
& x_{51} - x_{35} - x_{45} = +5 \\
& x_{63} - x_{26} = 0 \\
& x_{24} \leq 4 \\
& x_{ij} \geq 0, \forall (i, j) \in E
\end{aligned}$$

The optimal objective value is 56 (so there is an increase), and the optimal flow is:

$$x_{24} = 4, x_{26} = 6, x_{34} = 1, x_{51} = 5, x_{63} = 6$$

We see that the total transportation cost has increased to $Z^{**} = \$56$ (almost 7.7% increase, which is very bad for the company). This increase is caused by the road accident, preventing center 2 to send the 5 units of medicine directly to center 4 (it can only send 4 units of medicine). Center 2 must **split** the delivery of the medicine into two parts. The first part (4 units) is sent directly, that is

$$\text{center 2} \rightarrow \text{center 4} \quad (4 \text{ unit only}).$$

and the second part (1 unit) is sent via the path

$$\text{center 2} \rightarrow \text{center 6} \rightarrow \text{center 3} \rightarrow \text{center 4} \quad (1 \text{ unit only}).$$

(f) The dual of the MCF problem in part (b) is defined as

$$\begin{aligned}
\max \quad & -(-5w_1 + 10w_2 - 5w_3 - 5w_4 + 5w_5 + 0w_6) \\
& w_1 - w_2 + 2.5 \geq 0 \\
& w_2 - w_4 + 1.8 \geq 0 \\
& w_2 - w_6 + 3.1 \geq 0 \\
& w_3 - w_4 + 1.2 \geq 0 \\
& w_3 - w_5 + 2 \geq 0 \\
& w_4 - w_5 + 0.9 \geq 0 \\
& -w_1 + w_5 + 4 \geq 0 \\
& -w_3 + w_6 + 1.5 \geq 0
\end{aligned}$$

The optimal objective value is $\Omega^* = 52$, the same as the original MCF problem of part(b). This was expected because of Duality theory.

The optimal dual solution is

- $w_1 = 4$
- $w_2 = -2.7$
- $w_3 = 1.9$
- $w_4 = -0.9$
- $w_5 = 0$
- $w_6 = 0.4$

The dual variable w_i of a node i represent the marginal cost of supplying one extra unit at that node. If $w_i > 0$ then increasing the supply by one unit of medicine to it increases the total transportation cost by w_i . On the other hand, if $w_i < 0$ then adding a unit of medicine to node i decreases the total transportation cost by w_i . Lastly, if $w_i = 0$ then we will not observe any change in the total transportation cost by adding one unit of medicine to that center.

Note that center 4 has a dual variable that is negative (node 4 is a demand node). This means if we were to add one unit of medicine to that center (i.e., increase the available units from 15 to 16) the total transportation costs would decrease by \$0.9.

Remark: $w_i, i = 1, 2, \dots, 6$ are the dual variables (or Lagrange multipliers) associated with the i -th constraints of the MCF problem. The constraints have remarkable structure of the form

$$w_i - w_j + c_{ij} \geq 0, \quad \forall (i, j) \in E$$

Hence a compact form of the Dual of the MCF problem can be defined as

$$\begin{aligned}
\max \quad & -b^T w \\
& w_i - w_j + c_{ij} \geq 0, \quad \forall (i, j) \in E
\end{aligned} \tag{1}$$

2. *Problem 2 (35 points). Demand Prediction, Multi-Objective Least Squares, and Re-Optimization.*

Rural Lifeline Logistics has realized that the "Units Required" ($\mathbf{d}$), used in the redistribution model in Problem 1, are a critical source of uncertainty and may change in the future. To create a more robust logistics plan, the company has decided to use historical data to develop a predictive linear model for this quantity. The goal of this problem is to find a set of predictive coefficients ($\boldsymbol{\beta}$) and then use the resulting forecasts to re-optimize the medicine's transportation network.

The company uses a linear model to predict the required units of medicine for center i :

$$\hat{Y}_i \approx \beta_0 + \beta_1 X_{i,1} + \beta_2 X_{i,2} + \beta_3 X_{i,3} = \mathbf{X}\boldsymbol{\beta}$$

where $X_{i,j}$ are the features of the geographic region that center i serves, and $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3)^T$ is the vector of coefficients that needs to be computed. In particular, the features influencing demand are:

- $X_{i,1}$: Population of the geographic region of center i (in thousands of people, 'k')
- $X_{i,2}$: Patient Visits in center i (in hundreds of visits, 'h')
- $X_{i,3}$: Severity Index of cases in center i (it is a rating from 1 to 10)

The company seeks a single (robust) coefficient vector ($\boldsymbol{\beta}$) that minimizes the combined prediction error across the three past operational periods ($t = 1, 2, 3$).

Table 3 contains the data for the three key features for all six centers.

Center i	Population (k) $X_{i,1}$	Visits (h) $X_{i,2}$	Severity Index $X_{i,3}$
1	3.0	12	7.5
2	2.0	8	5.0
3	1.5	6	4.0
4	4.0	18	9.0
5	1.0	4	2.5
6	2.5	10	7.0

Table 3

Furthermore, Table 4 provides the historical observed demand ($\mathbf{Y}^{(t)}, t = 1, 2, 3$), corresponding to the d_i values in Problem 1, for each center across the three periods.

Center i	Demand Period 1 $Y_i^{(1)}$	Demand Period 2 $Y_i^{(2)}$	Demand Period P3 $Y_i^{(3)}$
1	15	16	13
2	10	9	12
3	10	8	10
4	20	18	21
5	5	7	6
6	15	16	15

Table 4

- (5 points) Formulate the Multi-Objective Least Squares problem in terms of the variable vector $\boldsymbol{\beta}$ (i.e., the coefficients of the linear model). This problem minimizes the combined sum of the squared prediction errors across all three periods.
- (5 points) Define the normal equations of the optimization problem in part (a)
- (10 points) Use Python to solve the normal equations (you can use `numpy` and `np.linalg.solve`).
 - Report the optimal coefficient vector $\boldsymbol{\beta}^* = [\beta_0, \beta_1, \beta_2, \beta_3]^T$.
 - Report the minimum combined Sum of Squared Errors (SSE).

- (d) (15 points) The company received the latest demographic and health data, shown in Table 5. This data represents the expected conditions for the upcoming operational period. For example, the increase in Population, Patient Visits, and Severity Index for Center 1 indicates an expected rise in medical need in that region, making a prediction of required demand ($\mathbf{d}$) necessary.

Center i	Population (k) $X_{i,1}$	Visits (h) $X_{i,2}$	Severity Index $X_{i,3}$
1	3.5	14	8.0
2	2.0	9	5.5
3	1.0	5	4.5
4	4.5	19	9.5
5	1.5	6	3.0
6	3.0	11	7.5

Table 5

- Use your calculated β^* from part (c) and the new feature data (Table 5) to predict the new raw demand vector $\mathbf{d}^{\text{pred}} = \mathbf{X}_{\text{new}}\beta^*$. Round the predicted demands to the nearest integer.
- Calculate the total predicted deficit and modify the supply vector $\mathbf{b}$. Show your work in the following four steps:
 - Calculate the total predicted demand D^{pred} .
 - Calculate the total deficit by subtracting the original total supply ($\sum b_i = 75$)

$$\Delta = D^{\text{pred}} - 75.$$

- The company will buy the total deficit but it can only assign it to a single center. For example, if the company assigns the deficit to center 6, the modified supply vector would be

$$\tilde{\mathbf{b}}^{(6)} = [10, 20, 5, 15, 10, 15 + \Delta]^T$$

and the net flow requirement vector would be $\mathbf{R}^{(6)} = \tilde{\mathbf{b}}^{(6)} - \mathbf{d}^{\text{pred}}$. Similarly for any other center $i = 1, 2, \dots, 6$. Determine which center is more economical to receive the entire deficit (that is, the center that will result in the lowest total transportation cost). You can use information and results from both Problem 1 and Problem 2. You must fully justify your answer.

Solutions

- (a) The multi-objective least squares is defined as the unconstrained optimization problem:

$$\min_{\beta} \sum_{k=1}^3 \|\mathbf{X}\beta - \mathbf{Y}^{(k)}\|_2^2$$

where

$$X = \begin{bmatrix} 1 & 3 & 12 & 7.5 \\ 1 & 2 & 8 & 5 \\ 1 & 1.5 & 6 & 4 \\ 1 & 4 & 18 & 9 \\ 1 & 1 & 4 & 2.5 \\ 1 & 2.5 & 10 & 7 \end{bmatrix}, \quad Y^{(1)} = \begin{bmatrix} 15 \\ 10 \\ 10 \\ 20 \\ 5 \\ 15 \end{bmatrix}, \quad Y^{(2)} = \begin{bmatrix} 16 \\ 9 \\ 8 \\ 18 \\ 7 \\ 16 \end{bmatrix}, \quad Y^{(3)} = \begin{bmatrix} 13 \\ 12 \\ 10 \\ 21 \\ 6 \\ 15 \end{bmatrix}$$

- (b) The objective function of the multi-objective least squares is

$$f(\beta) = \sum_{k=1}^3 \|\mathbf{X}\beta - \mathbf{Y}^{(k)}\|_2^2 = \|\mathbf{X}\beta - \mathbf{Y}^{(1)}\|_2^2 + \|\mathbf{X}\beta - \mathbf{Y}^{(2)}\|_2^2 + \|\mathbf{X}\beta - \mathbf{Y}^{(3)}\|_2^2$$

The optimality conditions are

$$\begin{aligned}\nabla f(\beta) = 0 &\implies 2X^T(X\beta - Y^{(1)}) + 2X^T(X\beta - Y^{(2)}) + 2X^T(X\beta - Y^{(3)}) \implies \\ 6X^T X\beta &= 3X^T(Y^{(1)} + Y^{(2)} + Y^{(3)}) \implies \beta = \frac{1}{3}(X^T X)^{-1}X^T(Y^{(1)} + Y^{(2)} + Y^{(3)})\end{aligned}$$

(c) The solution is

$$\beta^* = (1.8, -14.43, 2.283, 3.833)^T$$

and the Sum of Squares Error (SSE) is 19.4333.

(d) i. The new matrix X_{new} is defined as

$$X_{new} = \begin{bmatrix} 1 & 3.5 & 14.0 & 8.0 \\ 1 & 2.0 & 9 & 5.5 \\ 1 & 1.0 & 5.0 & 4.5 \\ 1 & 4.5 & 19.0 & 9.5 \\ 1 & 1.5 & 6.0 & 3 \\ 1 & 3.0 & 11.0 & 7.5 \end{bmatrix},$$

The predicted demand is

$$d^{\text{pred}} = X_{new}\beta^* = (13.9167, 14.5667, 16.033, 16.65, 5.35, 12.3667)^T$$

When we round it we obtain

$$d^{\text{pred}} = (14, 15, 16, 17, 5, 12)^T$$

ii. Total predicted demand:

$$D^{\text{pred}} = \sum_{i=1}^6 d_i^{\text{pred}} = 14 + 15 + 16 + 17 + 5 + 12 = 79$$

Total deficit: $\Delta = D^{\text{pred}} - 75 = 79 - 75 = 4$.

iii. We solve the MCF problem with the the new $\tilde{b}^{(i)}$ for all $i = 1, 2, \dots, 6$. This will require to re-optimize the original MCF six times (solving 6 LPs). It is computationally expensive process but it gives us all the flows for all these scenarios. We list the total transportation cost for each of these cases:

- Assigning Δ to center 1: total transportation cost = 77.4

$$(x_{12}, x_{24}, x_{26}, x_{34}, x_{35}, x_{45}, x_{51}, x_{63}) = (5, 2, 8, 0, 0, 0, 5, 11)$$

- Assigning Δ to center 2: total transportation cost = 67.4

$$(x_{12}, x_{24}, x_{26}, x_{34}, x_{35}, x_{45}, x_{51}, x_{63}) = (1, 2, 8, 0, 0, 0, 5, 11)$$

- Assigning Δ to center 3: total transportation cost = 49.0 (assign entire deficit here)

$$(x_{12}, x_{24}, x_{26}, x_{34}, x_{35}, x_{45}, x_{51}, x_{63}) = (1, 2, 4, 0, 0, 0, 5, 7)$$

- Assigning Δ to center 4: total transportation cost = 78.6

$$(x_{12}, x_{24}, x_{26}, x_{34}, x_{35}, x_{45}, x_{51}, x_{63}) = (3, 0, 8, 0, 0, 2, 7, 11)$$

- Assigning Δ to center 5: total transportation cost = 93.4

$$(x_{12}, x_{24}, x_{26}, x_{34}, x_{35}, x_{45}, x_{51}, x_{63}) = (5, 2, 8, 0, 0, 0, 9, 11)$$

- Assigning Δ to center 6: total transportation cost = 55.0

$$(x_{12}, x_{24}, x_{26}, x_{34}, x_{35}, x_{45}, x_{51}, x_{63}) = (1, 2, 4, 0, 0, 0, 5, 11)$$

Hence assigning the entire deficit to center 3 yields the lowest total transportation cost for the redistribution of the medicine.

iv. We can immediately deduce that center 4 has the smallest dual variable and therefore it will be be beneficial to assign the entire deficit to that center as it will yield the lowest transportation cost.

3. Problem 3 (35 points). Production choices

After several years of hard work and diligent planning the R&D department of the Optimal Products Inc. has developed three (3) new products that will change the future of the world. To avoid unnecessary diversification and cannibalization of the company's existing product line, the CEO has imposed the following restrictions.

- At most two (2) out of the three new products should be produced.
- Although all three new products can be produced in the three (3) factories the company owns, just one of these factories must be selected to produce the chosen new products.
- Products can only be manufactured in the chosen factories.

The number of hours of production time needed per unit of each product and the total available time in each factory are given in Table 6:

	Product 1 (hours/unit)	Product 2 (hours/unit)	Product 3 (hours/unit)	Available time (hours)
Factory 1	3	4	2	30
Factory 2	4	6	2	40
Factory 3	2	5	3	35

Table 6

The profit generated by selling one unit of each product and the maximum number of units the company is planning to manufacture are shown in Table 7:

	Product 1	Product 2	Product 3
Profit per unit (\$)	5	7	6
Max. # units	4	6	2

Table 7

The company wants to maximize its profit and at the same time satisfy all the restrictions posed to its production.

- (10 points) Formulate this problem as an integer optimization problem. Clearly define the decision variables, objective function and constraints.
- (10 points) Use CVXPY to solve this integer optimization problem. Write down the optimal solution and the optimal objective value (Z_1^*).
- (5 points) The CEO is considering relaxing the requirement "at most two products" to "all three products could be produced".
 - modify the integer optimization model in part (a) to reflect the new requirement.
 - Use CVXPY to solve the new optimization problem.
 - Write down the new optimal solution and maximum profit (Z_2^*).
 - Calculate $Z_1^* - Z_2^*$ and explain what it means for the company.
- (10 points) After consultations with the strategic planning department, the CEO ask the R&D department to include the following requirement: "if product 2 is selected then product 3 must also be selected for production".
 - Formulate the CEO's requirement as a conditional constraint and add it to the integer problem in Part (a).
 - Use CVXPY to find the optimal solution of the new problem.
 - Write down the selected factory or factories, the optimal product quantities, and the final maximum profit (Z_3^*).
 - Calculate $Z_1^* - Z_3^*$ and explain what it means for the company.

Solution:

- The decision variables we need are

- x_i representing the number of units of product $i = 1, 2, 3$ to be produced, for $i = 1, 2, 3$. Note that for each product x_i is restricted to take a specific number of **integer** values, $x_i \in \mathbb{Z}, i = 1, 2, 3$. For example, the units we can produced for product 1 must be in the set $\{0, 1, 2, 3, 4\}$, because we are only allowed to produce up to 4 units. Note also that we have $x_i \geq 0, \forall i$.
- y_j representing the decision whether product j is produced or not, for $j = 1, 2, 3$. That is, $y_j = 1$, if product j is produced; $y_j = 0$, otherwise.
- z_k representing the decision whether factory k is selected or not, for $k = 1, 2, 3$. That is, $z_k = 1$, if factory k is selected; $z_k = 0$, otherwise.

The objective of the company is to maximize the total profit

$$\max 5x_1 + 7x_2 + 6x_3$$

We have the following constraints

- At most two products can be produced by the company:

$$y_1 + y_2 + y_3 \leq 2$$

- If a product is not produced the number of units produced must be equal to zero:

$$x_i \leq My_i, i = 1, 2, 3$$

- Only one of the three factories must be selected:

$$z_1 + z_2 + z_3 = 1$$

- Only the constraints associated with the selected factory must be considered (relevant)

$$3x_1 + 4x_2 + 2x_3 \leq 30 + M(1 - z_1)$$

$$4x_1 + 6x_2 + 2x_3 \leq 40 + M(1 - z_2)$$

$$2x_1 + 5x_2 + 3x_3 \leq 35 + M(1 - z_3)$$

- The number of units to be produced, for each products, must be less than or equal to the maximum allowed quantity:

$$x_1 \leq 4, x_2 \leq 6, x_3 \leq 2$$

So we have the following integer optimization problem

$$\begin{aligned} \max \quad & 5x_1 + 7x_2 + 6x_3 \\ & y_1 + y_2 + y_3 \leq 2 \\ & x_i \leq My_i, i = 1, 2, 3 \\ & z_1 + z_2 + z_3 = 1 \\ & 3x_1 + 4x_2 + 2x_3 \leq 30 + M(1 - z_1) \\ & 4x_1 + 6x_2 + 2x_3 \leq 40 + M(1 - z_2) \\ & 2x_1 + 5x_2 + 3x_3 \leq 35 + M(1 - z_3) \\ & x_1 \leq 4, x_2 \leq 6, x_3 \leq 2 \\ & x_i \in \mathbb{Z}, i = 1, 2, 3 \\ & y_i \in \{0, 1\} \\ & z_i \in \{0, 1\} \end{aligned} \tag{2}$$

(b) Your Python code should produce: optimal objective 55, optimal solution $x^* = (4, 5, 0)^T$.

(c) We simply need to remove the constraint

$$y_1 + y_2 + y_3 \leq 2$$

from the original problem, or replace it with the relaxed constraints

$$y_1 + y_2 + y_3 \leq 3$$

Your Python code should produce: optimal objective 60, optimal solution $x^* = (4, 4, 2)^T$.

- (d) We want to express the conditional relation "if product 3 is selected then product 1 must also be selected for production". This is expressed by the constraint

$$y_2 \leq y_3$$

Your Python code should produce: optimal objective 54, optimal solution $x^* = (0, 6, 2)^T$.